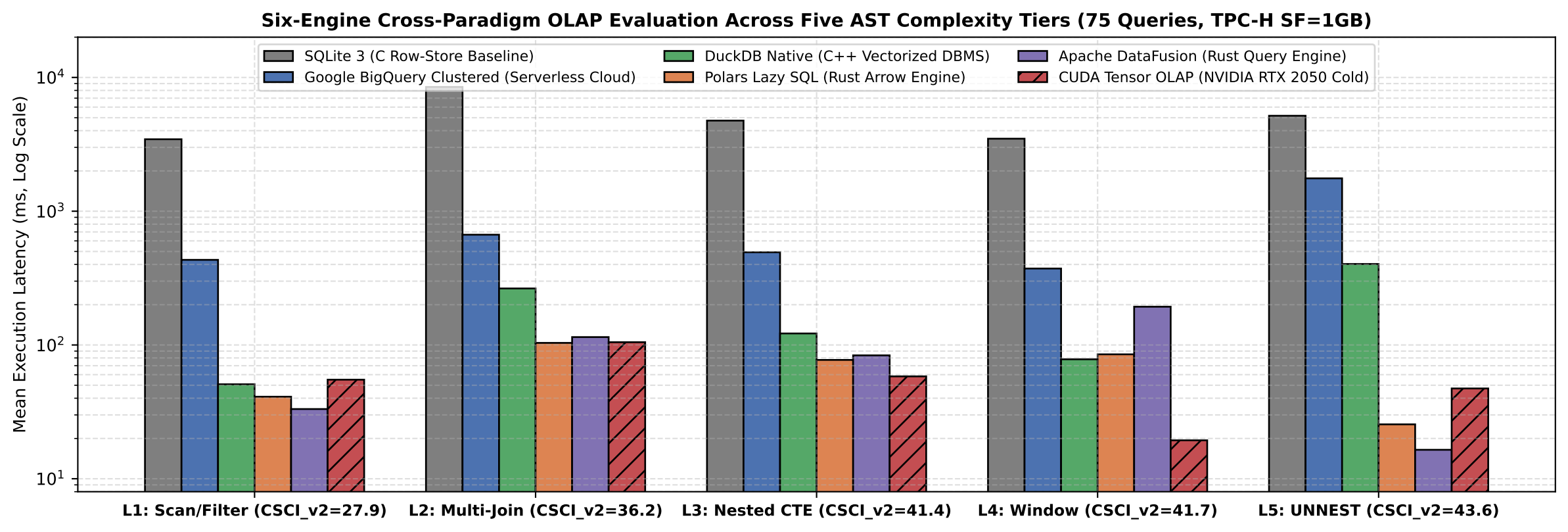
Six-Engine Cross-Paradigm OLAP Evaluation Across Five AST Complexity Tiers (75 Queries, TPC-H SF=1GB)
SQLite 3 (C Row-Store Baseline)
Google BigQuery Clustered (Serverless Cloud)
DuckDB Native (C++ Vectorized DBMS)
Polars Lazy SQL (Rust Arrow Engine)
Apache DataFusion (Rust Query Engine)
CUDA Tensor OLAP (NVIDIA RTX 2050 Cold)
Mean Execution Latency (ms, Log Scale)
10^4
10^3
10^2
10^1
L1: Scan/Filter (CSCI_v2=27.9)
L2: Multi-Join (CSCI_v2=36.2)
L3: Nested CTE (CSCI_v2=41.4)
L4: Window (CSCI_v2=41.7)
L5: UNNEST (CSCI_v2=43.6)